

# SE LAB 1

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QUESTION NO. 28 :

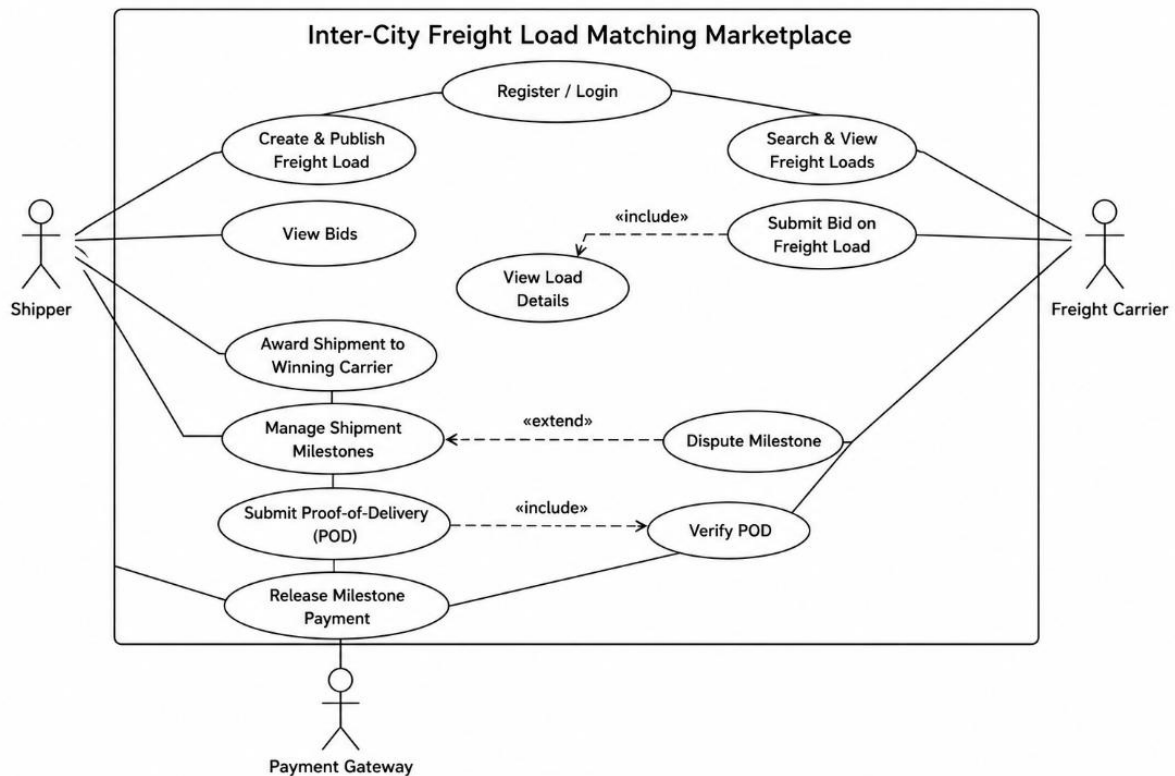
## Functional requirements (FR):

|   | A      | B          | C                                                                                                                                                                   | D        | E                                                                                                                                                                                                        | F                                                                                                           | G                                                                                                                                                         |
|---|--------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Req ID | Type       | Description                                                                                                                                                         | Priority | Acceptance Criteria (Pass/Fail)                                                                                                                                                                          | Rationale                                                                                                   | Comments                                                                                                                                                  |
| 2 | FR-001 | Functional | The system shall allow verified trucking carriers to submit competitive bids on posted freight shipments within a specified auction timeframe.                      | High     | <b>Pass:</b> Carrier bid is recorded and the shipper is notified.<br><b>Fail:</b> An unverified carrier or a carrier with invalid permits is allowed to submit a bid.                                    | Core functionality of the freight load matching marketplace and ensures only eligible carriers participate. | Verification should include carrier registration status, valid permits, and insurance confirmation.                                                       |
| 3 | FR-002 | Functional | The system shall allow shippers to create and publish freight load postings with cargo details, pickup and delivery locations, and auction timeframe.               | High     | <b>Pass:</b> A freight load containing all mandatory details is successfully published and becomes visible to eligible carriers.<br><b>Fail:</b> A load with missing mandatory information is published. | Allows shippers to make available freight loads visible to carriers for bidding.                            | Mandatory details should include cargo type, weight/volume, origin, destination, pickup window, delivery window, and auction end time.                    |
| 4 | FR-003 | Functional | The system shall allow shippers to view submitted bids and award a freight shipment to a selected winning carrier.                                                  | High     | <b>Pass:</b> The selected carrier is notified and the shipment status changes to "Awarded."<br><b>Fail:</b> The shipment is awarded without selecting a valid submitted bid.                             | Completes the load-matching process between the shipper and carrier.                                        | Only the shipper can award the shipment. Once awarded, bidding is closed for that shipment.                                                               |
| 5 | FR-004 | Functional | The system shall allow carriers to update shipment milestones and submit electronic proof-of-delivery (POD) for completed milestones.                               | High     | <b>Pass:</b> The milestone status and POD are successfully recorded with a timestamp.<br><b>Fail:</b> An invalid or missing POD is accepted for a completed milestone.                                   | Provides shipment tracking and supplies the proof required before payment release.                          | Milestones should include key stages such as pickup, in-transit, and delivery. POD should be in accepted formats, such as a photo with GPS and timestamp. |
| 6 | FR-005 | Functional | The system shall automatically release milestone payments to the carrier upon successful verification of the submitted proof-of-delivery according to escrow rules. | High     | <b>Pass:</b> Payment is released only after valid POD verification.<br><b>Fail:</b> Payment is released without successful POD verification.                                                             | Implements the automated milestone-payment functionality specified in the problem statement.                | Escrow rules should define payment amounts per milestone and disbursement conditions after verification.                                                  |

## Non functional requirements :

| 1 | Req ID  | Type          | Description                                                                                                                                                       | Priority | Acceptance Criteria (Pass/Fail)                                                                                                                                                                                   | Rationale (Short Justification)                                                                   |
|---|---------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 2 | NFR-001 | Nonfunctional | The system shall release milestone payments automatically within 60 seconds after successful electronic proof-of-delivery (POD) verification.                     | High     | <b>Pass:</b> 95% of verified POD transactions result in payment release within 60 seconds under peak load.<br><b>Fail:</b> Payment takes more than 60 seconds or is released without successful POD verification. | Ensures timely payment while maintaining the security of the escrow process.                      |
| 3 | NFR-002 | Nonfunctional | The system shall maintain at least 99.5% monthly availability for core freight bidding, shipment tracking, and payment services, excluding scheduled maintenance. | High     | <b>Pass:</b> Monthly monitoring shows availability $\geq 99.5\%$ .<br><b>Fail:</b> Monthly availability falls below 99.5%.                                                                                        | The marketplace needs reliable access because bidding and shipment milestones are time-sensitive. |

## UML USE CASE DIAGRAM:



## Use-Case Flow Specification:

### Use Case: Submit Bid on Freight Load

**Primary Actor:** Freight Carrier

**Supporting Actor:** Shipper

**Goal:** To allow a verified freight carrier to submit a competitive bid for a posted freight shipment within the specified auction timeframe.

#### Preconditions

1. The freight carrier is registered and verified in the system.
2. The carrier has valid permits required for bidding.
3. A freight load has been created and published by a shipper.
4. The auction for the freight load is still open.

### **Postconditions**

1. The carrier's valid bid is recorded in the system.
2. The bid is associated with the selected freight load.
3. The shipper is notified about the new bid.
4. The bid is available for the shipper to review.

### **Main Success Scenario**

1. Freight Carrier logs in to the system.
2. System authenticates the carrier and displays available freight loads.
3. Carrier searches for and selects an active freight load.
4. System displays the freight load details, including cargo information, pickup and delivery locations, and auction timeframe.
5. Carrier enters the bid amount and any required additional information.
6. System validates the carrier's verification status, permits, bid details, and auction status.
7. System records the valid bid with a timestamp.
8. System notifies the Shipper that a new bid has been submitted.
9. System displays confirmation of successful bid submission.
10. Use case ends successfully.

### **Alternate Flow – Bid Validation Failure**

#### **A1. Invalid Carrier or Bid Details**

1. At Step 6, the system detects an invalid condition, such as an unverified carrier, invalid permit, incomplete bid information, or an expired auction.
2. System displays an appropriate error message.
3. Carrier corrects the invalid information and resubmits the bid.
4. If validation succeeds, the flow continues from Step 7 of the Main Success Scenario.
5. If the carrier cannot satisfy the validation requirements, the bid is not recorded and the use case ends.